

SCS2310 - Digital Signal Processing

Convolution

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What is Convolution?

- **Definition:** A mathematical operation that combines two signals to form a third signal.
- **Notation:** $y[n] = x[n] * h[n]$
- **Importance:** It is the single most important technique in Digital Signal Processing (DSP).
- **Key Relationship:** It relates the three fundamental signals of interest in a linear system:
 - The Input Signal ($x[n]$)
 - The Impulse Response ($h[n]$)
 - The Output Signal ($y[n]$)

The Delta Function

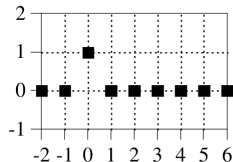
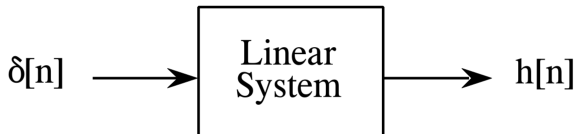
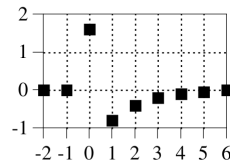
- **Symbol:** $\delta[n]$ (Greek letter delta)
- **Definition:** A normalized impulse.

$$\delta[n] = \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{if } n \neq 0 \end{cases}$$

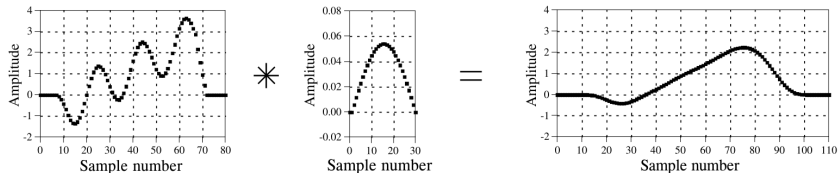
- Also known as the **unit impulse**.
- **Property:** Any discrete signal can be decomposed into a set of scaled and shifted delta functions.
 - Example: A sample of value -3 at index 8 is $-3\delta[n - 8]$.

The Impulse Response

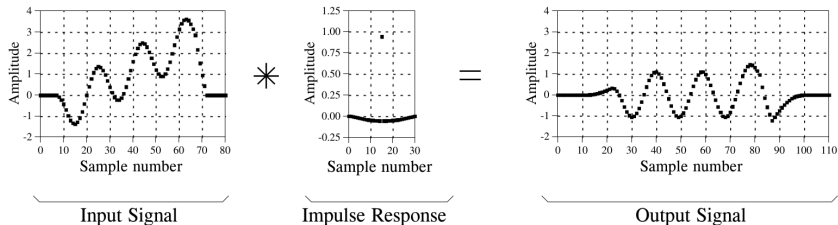
- **Symbol:** $h[n]$
- **Definition:** The signal that exits a linear system when a delta function ($\delta[n]$) is the input.
- **Significance:**
 - If you know a system's impulse response, you know everything about the system.
 - Different systems have different impulse responses.
- **Alternative Names:**
 - Filter kernel (in filtering)
 - Point spread function (in image processing)

Delta
FunctionImpulse
Response

a. Low-pass Filter



b. High-pass Filter



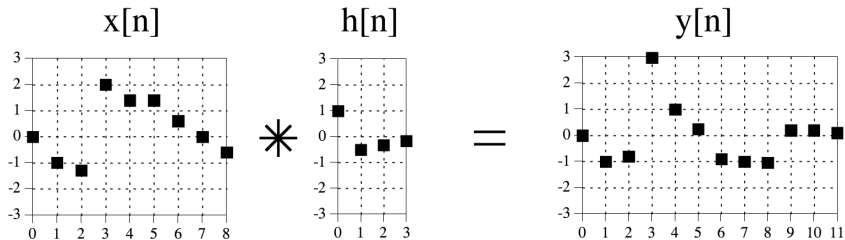
Conceptualizing Convolution

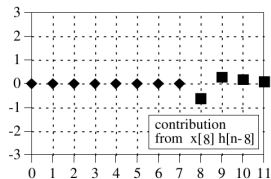
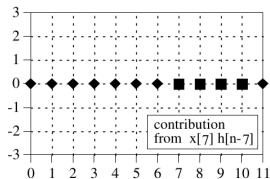
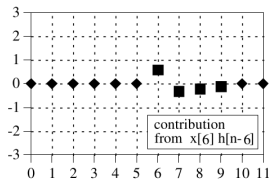
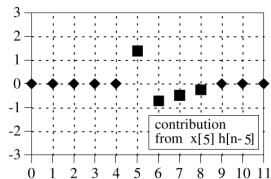
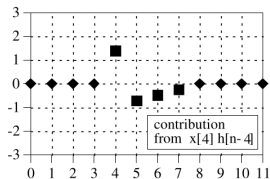
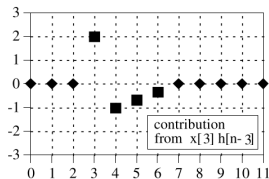
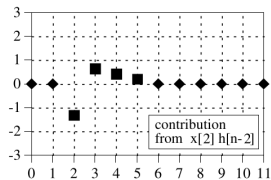
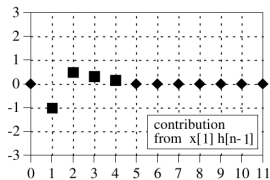
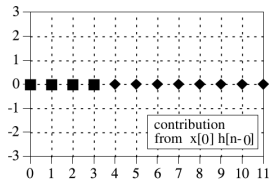
The Fundamental Concept of DSP:

- 1 Decompose the input signal into simple additive components (impulses).
- 2 Pass each component through the linear system individually.
- 3 Synthesize (add) the resulting output components to get the final signal.

The Input Side Algorithm

- This view analyzes how **each sample in the input signal** contributes to many points in the output signal.
- **Process:**
 - 1 Take an input sample $x[i]$ at position i .
 - 2 Scale the system's impulse response $h[n]$ by $x[i]$.
 - 3 Shift the scaled impulse response to position i .
 - 4 Add this scaled, shifted waveform to the output accumulator.
- **Result:** The output is a superposition of scaled and shifted impulse responses.

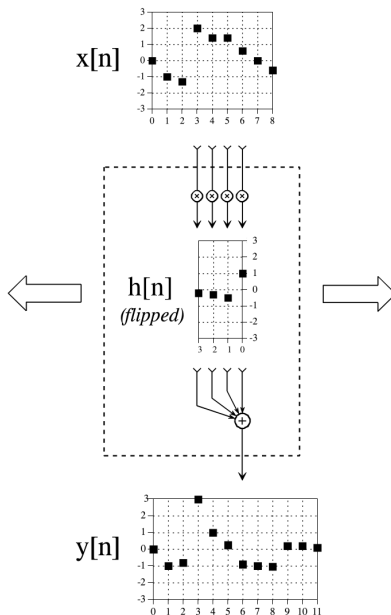


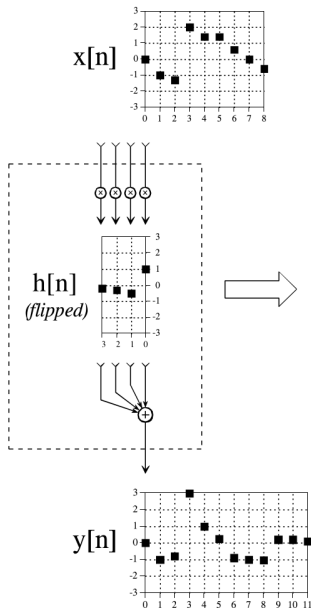
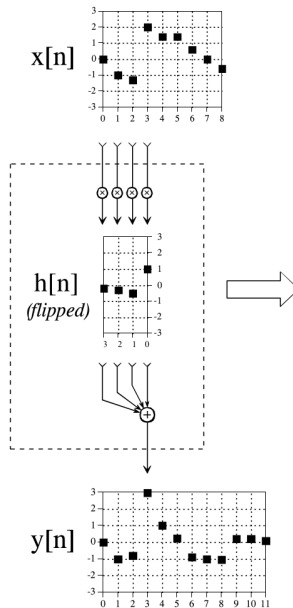


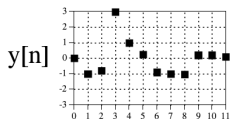
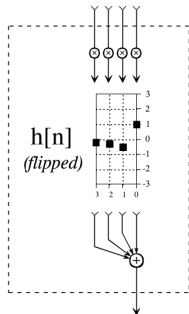
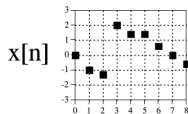
The Output Side Algorithm

- This view analyzes how **each sample in the output signal** is determined by contributions from many input points.
- This is how most DSP mathematical formulas and computer algorithms are written.
- **Mechanism:**
 - Imagine a "Convolution Machine" moving across the input signal.
 - To calculate a specific output point $y[n]$, we look at the input samples surrounding n .
 - These input samples are weighted (multiplied) by the impulse response points and summed.

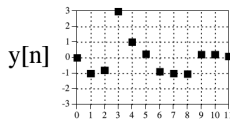
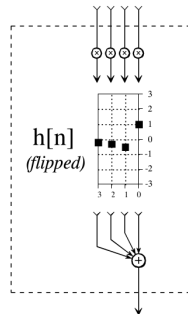
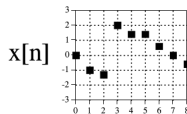





a. Set to calculate $y[0]$

b. Set to calculate $y[3]$



c. Set to calculate $y[8]$



d. Set to calculate $y[11]$

The Convolution Sum

The mathematical definition of convolution for discrete signals:

$$y[n] = \sum_{j=0}^{M-1} h[j] \cdot x[n-j]$$

- $y[n]$: Output signal sample at index n .
- $x[n]$: Input signal.
- $h[n]$: Impulse response of length M .
- j : Index variable for the impulse response.

Note: The input index is $n - j$, effectively "flipping" the impulse response relative to the input.

Length of Signals

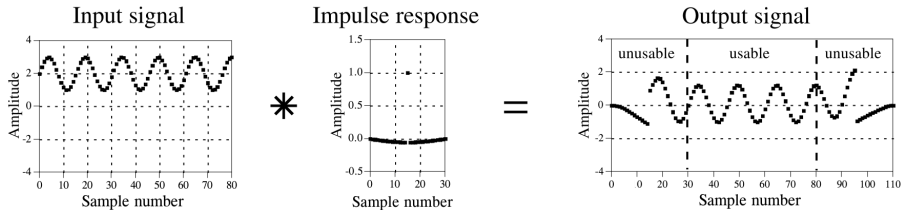
When convolving an input signal of length N with an impulse response of length M :

Output Signal Length

$$\text{Length}(y) = \text{Length}(x) + \text{Length}(h) - 1$$

$$L_y = N + M - 1$$

- Example: If input is 81 samples and impulse response is 31 samples, the output will be $81 + 31 - 1 = 111$ samples long.



Sum of Weighted Inputs

- The Output Side Algorithm can be viewed as a **Sum of Weighted Inputs**.
- Each sample in the output signal is equal to a sum of values from the input signal, where each input value is weighted by a coefficient.
- **Weighting Coefficients:** These are simply the values of the impulse response, $h[n]$.
- This perspective is very useful for understanding digital filters (e.g., Moving Average Filter).



Properties of Convolution

- A linear system's characteristics are completely specified by its **impulse response**.
- Convolution is the mathematical engine behind Linear Time-Invariant (LTI) systems.
- Applications include:
 - Digital Filters
 - Radar Detection
 - Echo Suppression
- We need to look at the following aspects of convolution:
 - 1 Common Impulse Responses
 - 2 Mathematical Properties
 - 3 Correlation
 - 4 Computation Speed

Common Impulse Responses

- 1 Identity (The Delta Function)
- 2 Amplification and Attenuation
- 3 Shift (Delay and Advance)
- 4 Echoes
- 5 Calculus-like Operations

1. Identity (The Delta Function)

- The simplest impulse response is the **Delta Function** $\delta[n]$.
- **Property:** Convolving any signal with a delta function results in the original signal.

$$x[n] * \delta[n] = x[n]$$

- Analogous to:
 - 0 in addition ($a + 0 = a$)
 - 1 in multiplication ($a \times 1 = a$)
- Used in systems that transmit or store signals without modification.

2. Amplification and Attenuation

- Formed by a **scaled delta function**.

$$h[n] = k\delta[n]$$

- **Operation:**

$$x[n] * k\delta[n] = kx[n]$$

- **Effects:**

- If $k > 1$: Amplification
- If $0 < k < 1$: Attenuation
- If $k < 0$: Inversion (flips the signal)

3. Shift (Delay and Advance)

- Formed by a **shifted delta function**.

$$h[n] = \delta[n - s]$$

- **Operation:**

$$x[n] * \delta[n - s] = x[n - s]$$

- Adds a shift equal to s samples.
- **Terminology:**
 - $s > 0$: Delay (shifts right)
 - $s < 0$: Advance (shifts left) - *Non-causal for real-time systems!*

4. Echoes

- Formed by summing a delta function and a scaled, shifted delta function.

$$h[n] = \delta[n] + k\delta[n - s]$$

- **Output:**

$$y[n] = x[n] + kx[n - s]$$

- Result: The original signal plus a scaled, delayed copy (echo).
- Used in audio effects, radar, and sonar analysis.

5. Calculus-like Operations

Convolution can mimic discrete differentiation and integration.

First Difference (Discrete Derivative):

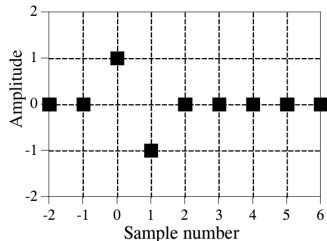
- $h[n] = \delta[n] - \delta[n - 1]$
- Operation: $y[n] = x[n] - x[n - 1]$
- Returns the slope of the signal.

Running Sum (Discrete Integral):

- $h[n] = u[n]$ (Unit Step Function: 1 for $n \geq 0$, 0 otherwise)
- Operation: $y[n] = \sum_{k=-\infty}^n x[k]$
- Inverse operation of the First Difference.

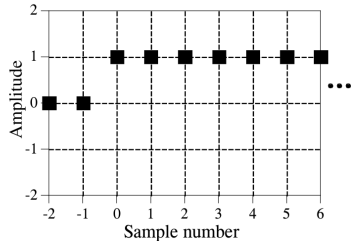
a. First Difference

This is the discrete version of the *first derivative*. Each sample in the output signal is equal to the *difference* between adjacent samples in the input signal. In other words, the output signal is the *slope* of the input signal.



b. Running Sum

The running sum is the discrete version of the *integral*. Each sample in the output signal is equal to the sum of all samples in the input signal to the *left*. Note that the impulse response extends to infinity, a rather nasty feature.



Mathematical Properties

- ① Commutative Property
- ② Associative Property
- ③ Distributive Property
- ④ Transference between Input and Output
- ⑤ Central Limit Theorem

1. Commutative Property

Definition

The order of convolution does not matter.

$$a[n] * b[n] = b[n] * a[n]$$

- **Implication:** In a linear system, the input signal and the impulse response can be swapped without changing the output.
- Mathematical symmetry, though physically distinct (signal vs. system).

2. Associative Property

Definition

The order of grouping in a series of convolutions does not matter.

$$(a[n] * b[n]) * c[n] = a[n] * (b[n] * c[n])$$

- **Application: Cascade Systems**

- If System 1 (h_1) is followed by System 2 (h_2):

$$y[n] = (x[n] * h_1[n]) * h_2[n]$$

- Equivalent to a single system with impulse response:

$$h_{total}[n] = h_1[n] * h_2[n]$$

- The order of cascaded LTI stages does not affect the final result.

3. Distributive Property

Definition

Convolution distributes over addition.

$$a[n] * (b[n] + c[n]) = (a[n] * b[n]) + (a[n] * c[n])$$

- **Application: Parallel Systems**
- Two systems (h_1, h_2) fed by the same input with outputs added.
- Equivalent to a single system with impulse response:

$$h_{total}[n] = h_1[n] + h_2[n]$$

4. Transference between Input and Output

- A linear change made to the input signal results in the *exact same* linear change in the output signal.
- Examples:
 - If $x[n]$ is doubled $\rightarrow y[n]$ is doubled.
 - If derivative of $x[n]$ is taken $\rightarrow$ output is derivative of $y[n]$.
 - If $x[n]$ is shifted $\rightarrow y[n]$ is shifted.
- Proven via Associative Property.

5. Central Limit Theorem

- The Central Limit Theorem states that a Gaussian distribution results when the observed variable is the sum of many random processes.
- The Central Limit Theorem has an interesting implication for convolution.
- **Concept:** Convolving a pulse-like signal with itself multiple times produces a signal resembling a **Gaussian** curve.
- The width of the Gaussian (σ) increases with each convolution:

$$\sigma_{total} = \sigma_{original} \times \sqrt{L}$$

where L is the number of convolutions.

- Explains why Gaussian distributions are common in nature (sum of many random events).

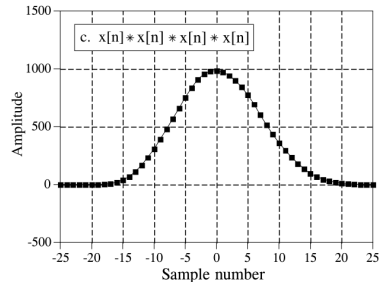
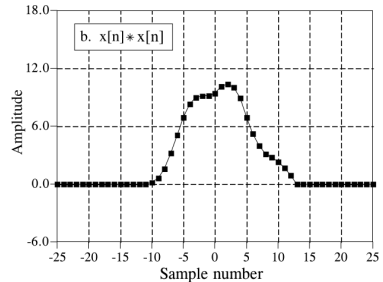
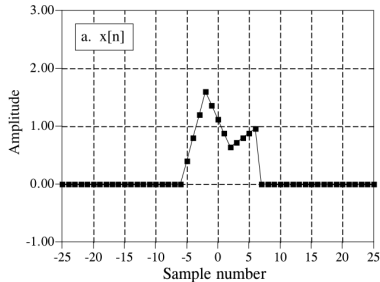


FIGURE 7-12

Example of convolving a pulse waveform with itself. The Central Limit Theorem shows that a Gaussian waveform is produced when an arbitrary shaped pulse is convolved with itself many times. Figure (a) is an example pulse. In (b), the pulse is convolved with itself *once*, and begins to appear smooth and regular. In (c), the pulse is convolved with itself *three* times, and closely approximates a Gaussian.

Correlation

- Correlation is a mathematical operation very similar to convolution.
- **Key Difference:**
 - **Convolution:** Flip the impulse response, then shift, multiply, and add.
 - **Correlation: No flipping.** Shift, multiply, and add directly.
- **Relation to Convolution:**
 - Correlation can be performed using convolution mathematics by pre-flipping one of the signals.
 - If Convolution is $a[n] * b[n]$, then Correlation is $a[n] * b[-n]$.

Types of Correlation

Cross-Correlation

- Correlation of two *different* signals.
- Used to determine if a specific target waveform (known signal) is present within a received signal.

Autocorrelation

- Correlation of a signal with *itself*.
- Used to find periodicities or repeating patterns within a signal.

Applications of Correlation

- **Matched Filtering (Optimal Detection):**

- The cross-correlation signal peaks where the target signal matches the received signal.
- It is the optimal linear method for detecting a known signal embedded in white noise.

- **Radar & Sonar:**

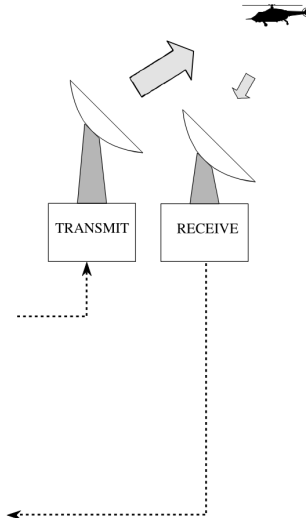
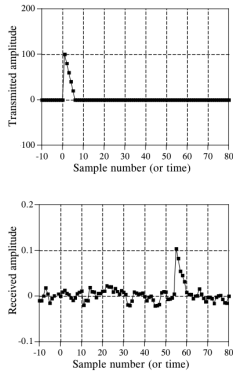
- A pulse is transmitted and an echo is received.
- Cross-correlation of the transmitted and received pulses reveals the time delay (distance) to the target.

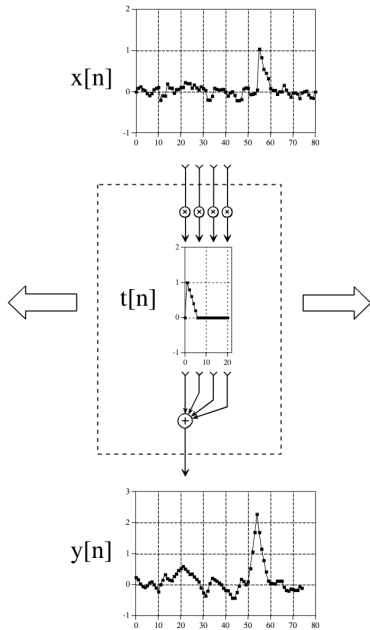
- **Image Processing:**

- Used to locate a small reference image within a larger scene.

FIGURE 7-13

Key elements of a radar system. Like other echo location systems, radar transmits a short pulse of energy that is reflected by objects being examined. This makes the received waveform a shifted version of the transmitted waveform, plus random noise. Detection of a known waveform in a noisy signal is the fundamental problem in echo location. The answer to this problem is *correlation*.





The Computational Burden

- Convolution is mathematically simple but computationally expensive.
- **Complexity:**
 - For an input signal of length N and an impulse response of length M .
 - Requires $N \times M$ multiply-accumulate operations.
- **Example:**
 - Signal (N) = 10,000 samples.
 - Impulse Response (M) = 100 samples.
 - Operations = $10,000 \times 100 = 1,000,000$ operations.
 - This can result in significant processing delays (latency).

Improving Speed

How do we handle the heavy computational load?

① Simplify the Data:

- Keep signals short.
- Use integer arithmetic instead of floating point where possible.

② Specialized Hardware:

- Digital Signal Processors (DSPs) are microprocessors optimized for single-cycle multiply-accumulate operations.

③ Better Algorithms (FFT Convolution):

- For long signals/kernels, standard convolution is inefficient.
- **FFT Convolution** reduces complexity significantly.
- Transforms convolution in time domain to multiplication in frequency domain.

